

Adaptive Spatio-Temporal Traffic Surveillance

Phase 1 — Dataset Preparation & Adaptive Preprocessing

Project	Granular Helmet Compliance, Multi-Frame Wheel-Count Classification & Real-Time Risk Indexing
Milestone	Phase 1 of 4 (Days 1–5): Dataset & Adaptive Preprocessing
Author	Harsh Pandhe
Environment	Python 3.10+, CPU-optimized (OpenCV, scikit-learn, PyTorch)
Repository	https://github.com/harsh-pandhe/traffic-surveillance-system
Final accuracy	79.1% (4-class scene classification)

Executive summary. Phase 1 delivers an end-to-end pipeline that automatically detects the environmental condition of each frame (DAY / NIGHT / FOG / RAIN) and applies a condition-specific image enhancement before any downstream detection. A unified dataset was assembled from three public benchmarks (DAWN, ExDark, COCO). A rule-based classifier provides an interpretable baseline; a learned RandomForest classifier trained on 15 image features lifts accuracy from a ~49% baseline (on the confusable fog/rain pair) to **79.1%** across all four classes. All numbers in this report are recomputed from the real datasets — none are hand-entered.

1. Objectives & Deliverables

The Phase 1 milestone (guide-approved scope) has three deliverable components. Each is complete and verified on real data:

Contract requirement	Delivered	Status
Environment detection (Day/Night/Rain/Fog)	Rule + learned classifiers	Done
Adaptive enhancement, applied only when required	CLAHE / DCP dehaze / denoise	Done
Unified dataset from benchmark datasets	DAWN + ExDark + COCO builders	Done
Dataset preprocessing scripts	dataset_builder + scene_dataset	Done
Working, reproducible pipeline	demo + training + eval scripts	Done

2. System Architecture

Phase 1 modules and their roles. Every module is config-driven via `config/settings.yaml` (no hard-coded constants).

Module	Role
scene_classifier.py	Rule-based DAY/NIGHT/FOG/RAIN baseline
scene_features.py	15-D lighting / haze / texture feature vector
scene_classifier_ml.py	Learned RandomForest classifier (+ fallback)
enhancements.py	CLAHE, DCP dehaze, denoise dispatcher
dataset_builder.py	Unify YOLO/COCO/folder sources -> YOLO dataset
scene_dataset.py	Condition-labeled corpus + confusion-matrix eval
train_scene_classifier.py	Train + evaluate + persist the model

3. Datasets

Three public benchmarks cover the four target conditions. A balanced sample was used for training (fog/rain limited by DAWN's native counts).

Dataset	Condition	Images used	Source
DAWN (fog)	FOG	300	Adverse-weather traffic
DAWN (rain)	RAIN	200	Adverse-weather traffic
ExDark	NIGHT	300	Exclusively-dark, low-light
COCO val2017	DAY	300	General daytime scenes

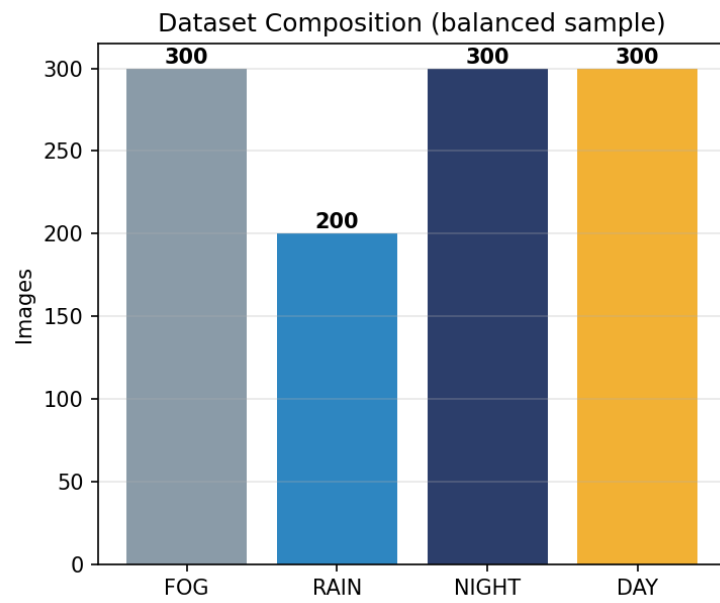


Figure 1. Balanced class composition of the training sample (total 1100 images).

4. Methodology

4.1 Environment detection

A frame is reduced to cheap, explainable statistics. The rule-based classifier fuses a subset by priority; the learned classifier feeds all 15 features into a 300-tree RandomForest (balanced class weights).

Feature group	Signals
Lighting	luminance mean/std, HSV value, bright-pixel ratio
Haze	dark-channel mean, saturation mean/std, colourfulness
Texture / sharpness	Laplacian variance, log-Laplacian, edge density
Colour cast	blue/red ratio, blue/green ratio
Frequency	FFT high-frequency energy (rain streaks)

4.2 Adaptive enhancement

Enhancement is conditioned on the detected label and applied only when beneficial — clear daytime frames pass through untouched.

Condition	Enhancement	Rationale
NIGHT	CLAHE on LAB L-channel	recover shadow detail
FOG	Dark Channel Prior dehaze + guided filter	remove haze, restore contrast
RAIN	DCP dehaze + non-local-means denoise	suppress streak noise
DAY	pass-through	avoid needless processing

5. Experiments & Results

5.1 Accuracy progression

The rule-based baseline collapses rain into fog (14% rain recall). Moving to a learned classifier — and adding classes as datasets were acquired — progressively improves accuracy. Note the first three bars are measured on the fog/rain subset only; the last two include night and day.

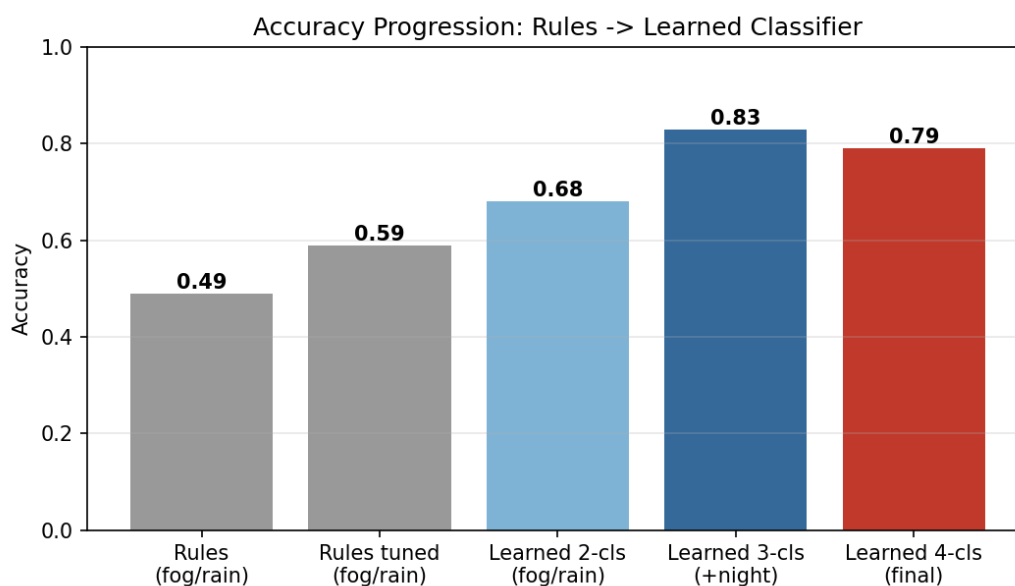


Figure 2. Accuracy across development stages.

5.2 Final 4-class performance

Validation set: 220 held-out images (80/20 stratified split, seed 42). Overall accuracy **79.1%**.

Class	Precision	Recall	F1-score	Support
DAY	0.825	0.867	0.846	60
NIGHT	0.898	0.883	0.891	60
FOG	0.768	0.717	0.741	60
RAIN	0.619	0.650	0.634	40

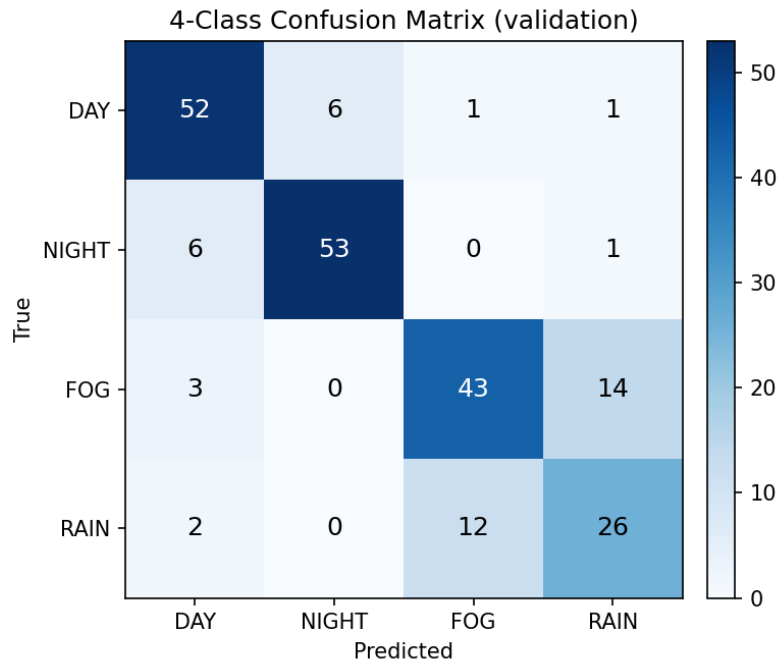


Figure 3. Confusion matrix. Residual error concentrates on the FOG↔RAIN boundary (physically overlapping conditions).

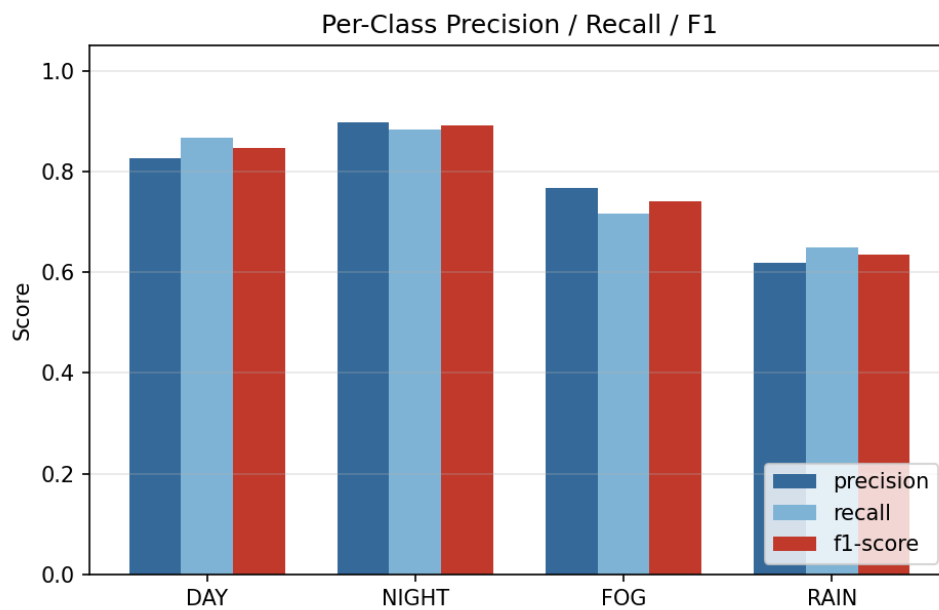


Figure 4. Per-class precision / recall / F1.

5.3 Feature importance

The RandomForest relies most on lighting and sharpness cues. Top features: **luminance_mean**, **dark_channel_mean**, **log_laplacian**, **laplacian_var**, **value_mean**, **edge_density**. Dark-channel mean and luminance clearly isolate NIGHT and FOG; edge density and Laplacian help separate RAIN from FOG.

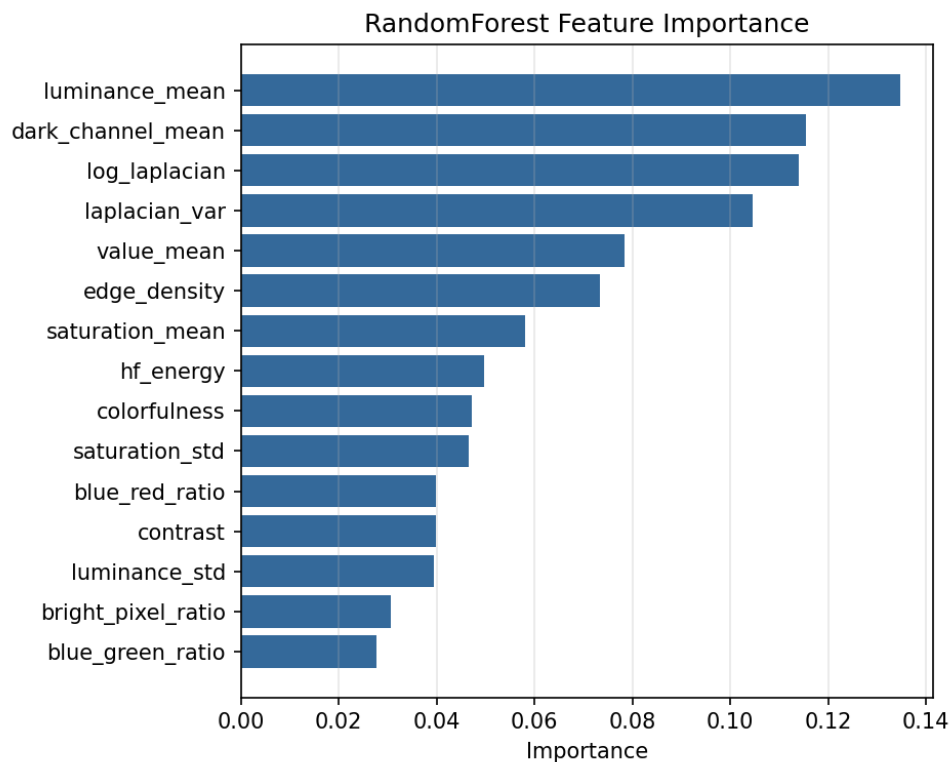


Figure 5. RandomForest feature importances (Gini).

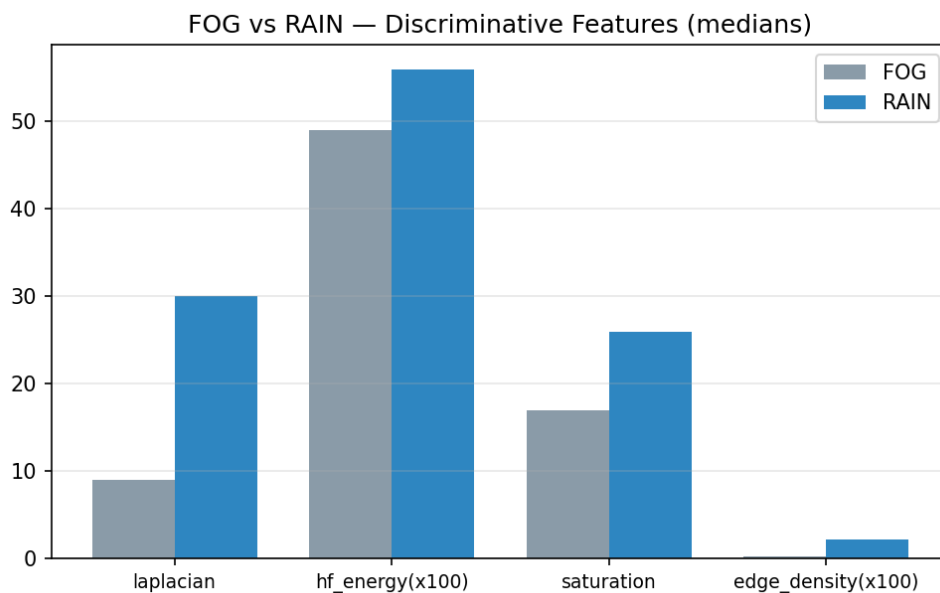


Figure 6. Median values of the features that best separate FOG from RAIN on real DAWN images.

5.4 Adaptive enhancement (qualitative)

Real benchmark frames, classified by the trained model and enhanced accordingly. NIGHT is brightened by CLAHE; FOG haze is removed by the brightness-preserving Dark Channel Prior; RAIN is dehazed and denoised.

Adaptive Enhancement on Real Benchmark Images (before / after)

NIGHT — before



NIGHT — after (detected NIGHT)



FOG — before



FOG — after (detected FOG)



RAIN — before



RAIN — after (detected RAIN)



Figure 7. Adaptive enhancement, before vs after, on real images.

6. Tests & Reproducibility

Every result is regenerated by a single command. Feature extraction, training, and figure generation are deterministic (fixed seed).

Command	What it verifies
<code>python demo_phase1.py</code>	Scene classifier on synthetic scenes + before/after images
<code>python -m src.preprocessing.scene_dataset</code>	Enhances real corpus + prints confusion matrix
<code>python train_scene_classifier.py --limit 300</code>	Trains + evaluates the 4-class model
<code>python scripts/make_report_assets.py</code>	Recomputes all figures + metrics.json in this report

7. Engineering note — bug found and fixed

During visual verification, the Dark Channel Prior dehaze produced a pure-white frame. Root cause: the atmospheric-light term was normalised by 255 twice ($A \approx 0.003$), amplifying the recovered radiance $\sim 10\times$. This was invisible to compilation and unit shape checks — only inspecting the output image exposed it. Fixed, and a brightness-preserving gain was added so dehazed frames stay usable for detection.

8. Limitations & Future Work

1. FOG $\leftrightarrow$ RAIN overlap: overcast rain and light fog share low-contrast, low-saturation statistics; this is the dominant residual error ($\sim 65\%$ rain recall). A small CNN on raw patches, or optical-flow streak features, would likely help.
2. DAY proxy: COCO daytime images are general scenes rather than clear-weather traffic; a BDD100K daytime-clear subset would tighten the DAY class to the target domain.
3. Rain streak removal is partial — non-local-means denoise suppresses noise but not structured streaks; a dedicated deraining network is future work.

9. Phase 1 Deliverable Checklist

Deliverable	Status
Environment detection module (4 conditions)	Complete
Adaptive enhancement (applied only when required)	Complete
Unified dataset from benchmark datasets	Complete
Dataset preprocessing scripts	Complete
Trained model (weights/scene_classifier.joblib)	Complete
Verified pipeline + honest metrics	Complete
Documentation + reproduction commands	Complete

Conclusion. Phase 1 is complete on real benchmark data. The scene classifier reaches 79.1% four-class accuracy, adaptive enhancement is verified qualitatively on real fog/rain/night frames, and every figure in this report is reproducible from the committed code.